

JIHUN KIM

C++, Python | C, C#, JavaScript, TypeScript | TensorFlow, Pandas, Git, Excel

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EDUCATION

The University of Tennessee, Knoxville

Knoxville, TN

Master of Science in Computer Science

Aug 2024 - May 2025

- GPA: 3.97/4.00
- Courses: Computer Systems Organization, Network Security, Algorithms, Natural Language Processing

Bachelor of Science in Computer Science

Aug 2020 - May 2024

Minor in Cybersecurity & Machine Learning

- GPA: 3.98/4.00 | Dean's List: Fall 2020 – Spring 2023, Spring 2024 (Summa Cum Laude)
- Courses: Data Structures & Algorithms, Systems Programming, Software Engineering, Applied Cryptography, Programming Languages and Systems, Deep Learning

EXPERIENCES

Machine Learning & Deep Learning Graduate Teaching Assistant

Knoxville, TN

University of Tennessee, Knoxville

Aug 2024 - Jul 2025

- Assisted students in debugging and implementing machine learning algorithms and *Python*-based assignments.
- Supported understanding of fundamental machine learning and deep learning concepts.
- Developed and graded assignments and quizzes to evaluate student understanding and reinforce learning objectives.

Science Undergraduate Laboratory Internships

Knoxville, TN

Oak Ridge National Laboratory

May 2023 - Jul 2023

- Developed NLP pipelines using BERT to improve interoperability between geographical categorization systems.
- Implemented information retrieval and semantic search methods for large-scale text datasets.
- Analyzed experimental results through data visualizations, including clustered bar charts and box plots.

Engineering Physics Undergraduate Teaching Assistant

Knoxville, TN

University of Tennessee, Knoxville

Aug 2022 - May 2023

- Supported students in computational modeling, data analysis, and problem-solving using Excel and MATLAB.
- Provided one-on-one technical assistance, troubleshooting coursework, and programming-related issues.

PROJECTS

NPC Decision-Making Systems, Team Project

Mar 2025 - May 2025

- Implemented a Behavior Tree-based NPC system using a modular hierarchical architecture for scalable AI design.
- Evaluated NPC decision-making algorithms including Utility AI, Behavior Trees, GOAP, and Finite State Machines using metrics such as implementation complexity, task effectiveness, and behavioral variability.

UTK Admission Chatbot, Team Project

Mar 2025 - Apr 2025

- Fine-tuned Meta's *LLaMA-3-8B* on a curated UTK admissions dataset to improve response accuracy.
- Improved response quality over baseline LLMs through hyperparameter tuning and prompt optimization.

Spiking Neural Network Drone, Team Project

Oct 2023 - May 2024

- Developed a *Spiking Neural Network*-based control system for autonomous Crazyflie drone navigation in simulation.
- Evaluated flight stability and control performance to support stable autonomous flight behavior.

ORGANIZATION

Hack4Impact

Sep 2023 - May 2024

- Developed a volunteer management and event coordination platform using *Next.js*, *TypeScript*, and *MongoDB*.
- Implemented API routes for creating, retrieving, and updating volunteer and event data.
- Built role verification and organization management interfaces, allowing administrators to manage volunteer permissions and maintain organizational information.

AWARDS

Engineering Scholarship, University of Tennessee, Knoxville

Aug 2021 - May 2024

- Awarded based on academic qualifications, extracurricular activities, and leadership roles.

Microsoft Excel Collegiate Challenge 2023 Certificate

Sep 2023 - Nov 2023

- Demonstrated proficiency in advanced Microsoft Excel concepts and data analysis techniques.

OLCF Hands-On HPC Certificate, Oak Ridge National Laboratory

Jun 2023 - Aug 2023

- Completed training in high-performance computing, GPU programming, and machine learning applications.